

Ingest Data From On-Premise SQL Server to Lakehouse

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1. What You Will Learn By the end of this session, you will understand:

- On-Premises SQL Server
- Why organizations use On-Premises SQL Server
- Challenges of accessing on-premises data from Microsoft Fabric
- On-premises Data Gateway
- Gateway installation and configuration
- Creating and testing an On-Premises SQL Server connection
- Full-load ingestion from SQL Server to Lakehouse
- Metadata-driven ingestion for multiple tables

2. What is On-Premises SQL Server?

On-Premises SQL Server means SQL Server is hosted within an organization's own infrastructure.

Key Points

- Server and database are managed by the organization
- Commonly used by enterprise applications
- Data remains inside the organization's network
- May support legacy and business-critical applications

Remember: On-premises = Database hosted inside the organization's own environment.

3. Why Organizations Still Use On-Premises SQL Server?

Common reasons:

- Legacy applications
- Existing infrastructure investments
- Business-critical applications
- Internal applications and systems
- Data governance requirements
- Security and network policies
- Applications not yet migrated to cloud

Key Point: Cloud adoption does not mean every operational database immediately moves to the cloud

4. Challenges of Accessing On-Premises SQL Server from Fabric

Microsoft Fabric is a cloud platform, while SQL Server may exist inside a private corporate network.

Common Challenges

- SQL Server may not be publicly accessible

- Network restrictions
- Firewall restrictions
- Authentication requirements
- Secure connectivity requirements
- Corporate security policies

Main Question How can Fabric securely access the on-premises SQL Server?

5. What is On-Premises Data Gateway?

The On-premises Data Gateway provides connectivity between on-premises data sources and Microsoft cloud services.

Gateway Responsibilities

- Provides connectivity to on-premises data
- Acts as a bridge between local environment and cloud
- Enables Fabric to access configured on-premises data sources

Important

Gateway Database The Gateway is a connectivity component, not the source database or destination.

6. Why Do We Need the Gateway?

The Gateway helps us:

- Connect on-premises data sources to Fabric
- Work with private/internal data sources
- Enable cloud-based data ingestion

Remember

On-Prem SQL Server → Gateway → Microsoft Fabric

The Gateway provides the connectivity path between the on-premises environment and Fabric.

7. Full Load What is Full Load?

Full Load means reading the required data from the source and loading it into the Lakehouse.

8. Metadata-Driven Ingestion

Instead of creating separate Copy Activities for every table, maintain table information in a configuration table.

Example Configuration

src_schema	src_table	sink_path	is_active
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saleslt	customer	global/customer	1
saleslt	product	global/product	1
dbo	emp	global/emp	1

Pipeline Pattern

Lookup



ForEach



Copy Customer

Copy Product

Copy Employee

The configuration is retrieved using:

SELECT *

FROM dbo.tblconfig

WHERE is_active = 1;

The Lookup output can then be used by the ForEach activity.

1. What does an on-premises SQL Server mean?

- A. SQL Server hosted only in Microsoft Fabric
- B. SQL Server hosted within an organization's own infrastructure
- C. SQL Server hosted only in Azure
- D. SQL Server hosted in a public cloud

- 2. Why do organizations continue using on-premises SQL Server?**
- A. Existing legacy and business-critical applications
 - B. To eliminate all security policies
 - C. To avoid using databases
 - D. To automatically migrate all applications to Fabric
- 3. What is one of the main challenges when accessing on-premises SQL Server from Microsoft Fabric?**
- A. Fabric cannot create pipelines
 - B. SQL Server may be inside a private corporate network
 - C. Lakehouse cannot store data
 - D. SQL Server cannot contain tables
- 4. What is the primary purpose of the On-premises Data Gateway?**
- A. To store data permanently
 - B. To replace SQL Server
 - C. To provide connectivity between on-premises data sources and Microsoft cloud services
 - D. To create Lakehouses
- 5. Which statement about the On-premises Data Gateway is correct?**
- A. It is the destination database
 - B. It is a connectivity component
 - C. It replaces the Fabric Pipeline
 - D. It is a Lakehouse storage layer
- 6. What is the purpose of using a Gateway when connecting on-premises data to Fabric?**
- A. To expose the SQL Server directly to the internet
 - B. To connect private/internal data sources to Fabric
 - C. To delete the source database
 - D. To convert SQL Server into a Lakehouse
- 7. In a metadata-driven ingestion pipeline, which activity is used to retrieve the active metadata records?**
- A. ForEach
 - B. Lookup
 - C. Copy
 - D. Delete
- 8. Which activity is typically used to process multiple metadata records dynamically?**
- A. Lookup
 - B. ForEach
 - C. Notebook
 - D. Trigger
- 9. In the metadata-driven example, what does @item().src_table represent?**
- A. The destination Lakehouse path
 - B. The source table name
 - C. The pipeline name
 - D. The Gateway name
- 10. What is the purpose of the is_active column in the metadata configuration?**
- A. To identify whether a mapping should be processed
 - B. To store the SQL Server password
 - C. To specify the Gateway name

D. To identify the Lakehouse workspace